MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5896

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Double diode, subminiature, 400 MHz

Outline	---	3-1(EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2a	2k	h	sd	1a	h	1k	nc

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Epp/a	epx	Ehk	R1	C1	Io/a	ib/a	i surge/a	TE	Alt
Unit:	V	Vac	v	v	Ohms	µF	mAdc	ma	ma	°C	ft
Maximum:	6.6	---	460 (see note 1)	360 (see note 1)	---	---	10	60	350	+220	note 8
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	165	---	0	11,000	8	---	---	---	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	280	320	mA
1353	Operation of rectifiers	See notes 3 and 6	0.4	II	Io	16	---	mAdc
1336	Heater-cathode leakage	Ehk = 360 Vdc (see note 7)	0.4	II	Ihk	---	20	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulaton of electrodes	E(a to all) = -300 Vdc (see note 7)	2.5	I	---	---	---	---
1031	Variable-frequency vibration	No voltages; post shock and vibration-fatigue test end points	10.0	See note 4	---	---	---	---
1231	Emission	Eb = 10 Vdc (see note 7)	2.5	I	Is	30	---	mAdc
1256	Electrode current	Ebb = 0; Rp = 40,000 ohms (see note 7)	2.5	I	Ib	5.0	25	µAdc
1256	Electrode current (anode) (difference between sections)	Ebb = 0; Rp = 40,000 ohms	2.5	I	ΔIb	---	5.0	µAdc
1331	Direct-interelectrode capacitance	0.405 in. dia shield	6.5	Code F	C	---	0.026	pF
		1a to 2a			C	2.5	3.5	pF
		1a to h+1k+sd			C	2.5	3.5	pF
		2a to h+2k+sd			C	3.5	4.9	pF
		1k to h+1a+sd			C	3.5	4.9	pF
		2k to h+2a+sd			C			
1116	Lead fatigue		2.5	Code F	---	---	---	---
2126	Envelope stain		4.0	I	---	---	---	---
1041	Shock	450 G; Ehk = +100 Vdc	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 4	---	---	---	---
---	Post-shock and vibration-fatigue test end points:							
1336	Heater-cathode leakage	Ehk = 360 Vdc	---	---	Ihk	---	60	µAdc
1353	Operation of rectifiers		---	---	Io	14	---	mAdc
1105	Permanence of marking		---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac; E1b = E2b = 0	---	---	---	---	---	---	---	---
1516	Stability life	TA = room (see note 2)	---	---	---	---	---	---	---	---
---	Stability life-test end point:									
1353	Change in operation of rectifiers of individual tubes		---	---	---	---	ΔIo t	---	10	%
1501	Intermittent life	Stability life-test conditions; TE = 220°C (min) (see note 5)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1301	Heater current		---	---	1	3	If	276	328	mA
1353	Operation of rectifiers		---	---	1	3	Io	14	---	mAdc
1353	Change in operation of rectifiers of individual tubes		---	---	1	3	ΔIo t	---	15	%
1336	Heater-cathode leakage	Ehk = 360 Vdc	---	---	1	3	Ihk	---	40	µAdc
1211	Insulation of electrodes	E(a to all) = -300 Vdc	---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points (1,000 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1301	Heater current		---	---	1	3	If	276	328	mA
1353	Operation of rectifiers		---	---	1	3	Io	14	---	mAdc
1353	Change in operation of rectifiers of individual tubes		---	---	1	3	ΔIo t	---	15	%
1336	Heater-cathode leakage	Ehk = 360 Vdc	---	---	1	3	Ihk	---	40	µAdc
1211	Insulation of electrodes	E(a to all) = -300 Vdc	---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.

NOTES:

1. The maximum voltage appearing between any pair of leads shall be no greater than the maximum peak inverse anode voltage rating.
2. In a full-wave life-test circuit, the values specified for R1 and C1 may be considered approximate and shall be adjusted initially to give not less than Io = 18 mAdc and ib = 50 mA with a bogey tube. (see note 6).
   Ehk = Eo + 117 Vac.
3. This test to be performed at the conclusion of the holding period.
4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
5. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Etd (±10 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
6. In a full-wave circuit, adjust Zp/a so that a bogey tube gives Io = 18 mAdc. A bogey tube has a tube voltage drop of Etd = 10 Vdc at Is = 50 mAdc per anode.
   Ehk = Eo + 117 Vac.
7. Test each unit separately.
8. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army -
Air Force - 17, 99
DLA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3179)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.